

Literature Status for arXiv:1208.2384: Adversarial Gowers Games

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2026-07-18

Status

This is a `literature_already_answered` packet. It records that Problem 1.2 in Christian Rosendal's *Determinacy of adversarial Gowers games*, arXiv:1208.2384, is explicitly answered by Noe de Rancourt's *Ramsey theory without pigeonhole principle and the adversarial Ramsey principle*, arXiv:1805.04954.

Source Question

Rosendal's Theorem 1.1 proves the adversarial Gowers-game dichotomy for $F_{\sigma\delta}$ and $G_{\delta\sigma}$ subsets of E^ω . On PDF page 3, immediately after Theorem 1.1, Problem 1.2 asks whether Theorem 1.1 holds for all Borel sets $\mathcal{A}$, or even for analytic sets in the presence of large cardinals.

Supporting Answer

The supporting paper explicitly identifies the same questions. On PDF page 6, de Rancourt states Rosendal's Question 1.10, asking whether every Borel set is adversariially Ramsey, and Question 1.11, asking whether every analytic set is adversariially Ramsey in the presence of large cardinals. The same paragraph says that the paper proves the answer to Question 1.10 is yes, and that the answer to Question 1.11 is yes in the presence of a measurable cardinal.

The Borel answer is Theorem 2.4, the abstract adversarial Ramsey principle: every Borel subset of X^ω is adversariially Ramsey. Immediately after the theorem, the paper says that, in the Rosendal space, the adversarial Gowers games defined there are exactly the games from the introduction, so Theorem 2.4 gives a positive answer to Question 1.10.

For the analytic large-cardinal clause, Corollary 2.6 says that if every Γ -subset of $\mathbb{R}^\omega$ is determined, then every Γ -subset of X^ω in an analytic Gowers space is adversariially Ramsey. The following paragraph invokes Martin's measurable-cardinal theorem for analytic determinacy and concludes that every analytic set in an analytic Gowers space is adversariially Ramsey; it explicitly says this answers Question 1.11.

Scope

This is not a new proof packet. It records a known positive literature answer to Rosendal's Problem 1.2. The Borel part is answered in ZFC through Borel determinacy. The analytic part is answered under the measurable-cardinal hypothesis used by de Rancourt, matching Rosendal's large-cardinal formulation but not giving a ZFC analytic result.

The supporting paper uses its own convention for naming the adversarial games in the introductory reformulation. For that reason, the identification here is based on de Rancourt’s explicit Rosendal Questions 1.10 and 1.11 and his stated positive answers, rather than on an isolated comparison of the letters A and B .

Search Evidence

The lane-1 queue selected arXiv:1208.2384. Cheap run indexes were searched for the arXiv id, source title, “adversarial Gowers games”, “strategic Ramsey”, “block sequences”, and “Banach game”; no duplicate packet was found. Targeted web search for the exact source title and for “adversarial Gowers games Borel analytic large cardinals” identified arXiv:1805.04954. The local source TeX and extracted PDF text verify the exact problem statement, the supporting paper’s explicit Rosendal-question restatement, Theorem 2.4, and the measurable-cardinal paragraph answering the analytic clause.

References

- [1] C. Rosendal, *Determinacy of adversarial Gowers games*, arXiv:1208.2384; Fund. Math. **227**, 163–178 (2014).
- [2] N. de Rancourt, *Ramsey theory without pigeonhole principle and the adversarial Ramsey principle*, arXiv:1805.04954; Trans. Amer. Math. Soc. **373**, 5025–5056 (2020).